

Juan David Muñoz-Bolaños

Rechenauerstraße 42, Innsbruck, Austria (Tirol)

+43 676 9115145

juan.munoz@i-med.ac.at

MED-EL

Hiring Manager

Fürstenweg 77

6020 Innsbruck, Austria

Subject: Application for Quality Engineer (m/f/d) - Ref. RDSY_22601

Dear Hiring Manager,

As an engineer with a strong background in R&D and high-precision metrology, I am highly motivated to transition my expertise into Quality Engineering at MED-EL. I have spent the last four years in Innsbruck developing complex optical systems where precision, reliability, and rigorous documentation were not just requirements, but the foundation of my work.

During my Ph.D. at the Medical University of Innsbruck, I managed the full lifecycle of advanced optical setups. This included technical specification, risk assessment of experimental configurations, and hands-on sourcing and qualification of components from international vendors. My experience in coordinating multidisciplinary projects and maintaining detailed technical documentation aligns closely with the change management and post-market support responsibilities at MED-EL. I am adept at bridging the gap between innovative R&D and the structured requirements of a quality management system.

My work style is characterized by conscientiousness and a detail-oriented approach. I am familiar with the high standards of the medical research environment and am eager to apply this mindset to ensure compliance with EU MDR and ISO 13485 standards. I am a proactive problem-solver who thrives in fast-paced environments and values the importance of traceability and continuous process improvement.

With my Ph.D. concluding in May, I am ready to bring my analytical skills and dedication to MED-EL. I am fluent in English and have a certified B2 level in German, allowing me to collaborate effectively in your international team. I look forward to the opportunity to contribute to the high quality and reliability of MED-EL's life-changing hearing solutions.

Sincerely,

Juan David Muñoz-Bolaños

Quality and R&D Engineer

Medical University of Innsbruck

Juan David Muñoz-Bolaños

Quality and R&D Engineer

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PROFESSIONAL SUMMARY

Engineer with a specialized background in high-precision metrology, R&D integration, and technical systems design. Proven expertise in managing complex engineering projects, from initial specification to final qualification. Highly detail-oriented with a strong commitment to reliability and documentation, eager to apply a systematic approach to Quality Engineering and regulatory compliance (ISO 13485/MDR) in the medical device industry.

TECHNICAL COMPETENCIES

Quality Engineering and Compliance

Strong focus on technical documentation, traceability, and risk management within R&D environments. Eager to apply engineering principles to EU MDR and ISO 13485 compliance.

System Integration and R&D Support

Experienced in the full lifecycle of complex technical systems, including change control, alternative part qualification, and V&V planning.

Technical Sourcing and Supplier Management

Proven track record of managing supplier relationships (e.g., Thorlabs, Edmund Optics) and ensuring component quality for high-precision instrumentation.

Analytical Problem Solving

Systematic approach to identifying and resolving technical issues, with a focus on continuous process improvement and operational efficiency.

PROFESSIONAL EXPERIENCE

Medical University of Innsbruck

Innsbruck, Austria

PhD Researcher - Systems and Quality Integration

Jan 2022 – Present

- **Technical Documentation and Risk Management:** Managed the end-to-end development of advanced microscopy setups, ensuring meticulous documentation and risk assessment of high-precision optical and electronic configurations.
- **Change Management and Qualification:** Oversaw product updates and hardware modifications, implementing verification and validation protocols to ensure system reliability and performance stability.
- **Supplier Quality and Sourcing:** Actively managed relationships with key hardware vendors, performing technical qualification of alternative components and ensuring adherence to strict performance specifications.
- **Cross-functional Support:** Collaborated with interdisciplinary teams to translate research requirements into robust, qualified technical solutions, feeding experimental learnings back into system improvements.

Leibniz Institute of Photonic Technology (IPHT)

Jena, Germany

R and D Researcher (Photonics)

Jun 2020 – Dec 2021

- **Prototyping and Calibration:** Designed and qualified fiber probe spectrometers; implemented rigorous calibration protocols to ensure measurement accuracy for clinical tissue studies.
- **Process Improvement:** Streamlined data analysis workflows using machine learning, improving the efficiency and reliability of experimental characterization.

INTECOL S.A.S.

Medellín, Colombia

Junior Machine Vision System Engineer

Mar 2018 – Mar 2019

- **Industrial Quality Control:** Implemented vision inspection systems for high-speed manufacturing, focusing on defect detection and automated quality assurance to minimize production errors.
- **Technical Specification:** Managed the selection and qualification of cameras, optics, and lighting systems, ensuring they met the demanding reliability standards of industrial production lines.

EDUCATION

Medical University of Innsbruck
Ph.D. in Adaptive Optics & Microscopy

Austria
Jan 2022 – Present

Friedrich Schiller University Jena
M.Sc. in Photonics

Germany
2019 – 2021

University of Cauca
B.Sc. in Physics Engineering

Colombia
2012 – 2018

PUBLICATIONS & AWARDS

- **Awards and Selections:** Selected for ASML and ZEISS Summer Schools; awarded COLFUTURO Grant 2026 for best Colombian professionals.
- Muñoz-Bolaños, J. D., et al. *Applied Sciences*, 14(11), 4726 (2024).
- Muñoz-Bolaños, J. D. et al. *ACS Photonics* 12(1), 176–184 (2025).

LANGUAGES

English (Fluent / Professional),
German (B2 - Working Proficiency),
Spanish (Native)

REFERENCES

Prof. Monika Ritsch-Marte
Head of Institute
monika.ritsch-marte@i-med.ac.at

Medical University of Innsbruck

AGREEMENT FOR GUESTS

to carry out a practical training
with

Mr. **Juan David Munoz Bolanos** Date of birth: **1994-06-04**
temporarily living at **Emil-Wölk-Str. 7, 07747 Jena** (guest),

Student of the Friedrich-Schiller-University Jena, Abbe School of Photonics, gets in execution of his practical course in the research department Spectroscopy and Imaging, working group Raman and Infrared Histopathology (0104) from 2020-06-02 up to 2020-12-31 admission to the necessary rooms within Leibniz-IPHT.

The use of further devices, media and other infrastructure takes place in agreement with the Heads of the respective departments.

Mr. Juan David Munoz Bolanos has to observe the security regulations for working and other regulations valid in Institute as well as instructions of the laboratory and clean room responsible persons of the Leibniz-IPHT.

Mr. Juan David Munoz Bolanos has to treat all affairs and procedures strictly confidential. Copies, duplicates and others of business and technical information of each kind are forbidden, unless these are sanctioned by the responsible Department Heads.

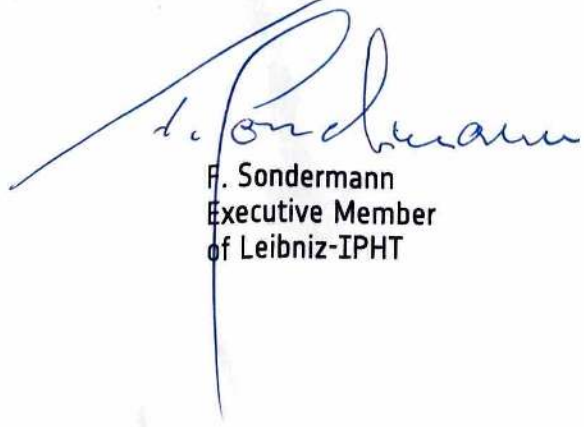
Publications, reports and lectures about topics which belong together with the fields of work of Leibniz-IPHT require the previous written agreement of the Executive Board.

The liability of Mr. Juan David Munoz Bolanos is limited to cases of wilful action and gross negligence.

Mr. Juan David Munoz Bolaos is insured against accidents on the Friedrich-Schiller-University Jena, Abbe School of Photonics in the context of his/her practical course during his/her length of stay at Leibniz-IPHT.

Jena, 18 May 2020


Prof. Dr. Popp
Chairman
of Leibniz-IPHT


F. Sonderrmann
Executive Member
of Leibniz-IPHT


Juan David Munoz Bolanos
Guest

INTECOL

OPTIMIZING THE RHYTHM OF THE WORLD

Medellin, February 4, 2019

TO WHOM IT MAY CONCERN

Integración Tecnológica Industrial Intecol S.A.S. with Tax ID (NIT) 811.041.410-4 certifies that Mr. JUAN DAVID MUÑOZ BOLAÑOS, identified with Citizenship ID No. 1.061.772.278, has been employed by this company since July 3, 2018, under an indefinite term contract, serving in the position of Junior Project Engineer;

earning a salary of One million five hundred thousand pesos (\$1,500,000) plus three hundred thousand pesos (\$300,000) mobility allowance.

I will gladly address any additional information at the phone number 316 3322.

We appreciate your attention,

INTECOL

NIT. 811.041.410-4 - Tel.: 3163822

LENY ADRIANA RUIZ PINO

Administrative and Financial Director

Integración Tecnológica Industrial Intecol S.A.S.
www.intecol.com.co
Medellin Calle 42 #63c-82
PBX. (574) 316 3322 Cell: (57) 301 430 2532



URKUNDE

Die Friedrich-Schiller-Universität Jena verleiht durch die
Physikalisch-Astronomische Fakultät
mit dieser Urkunde

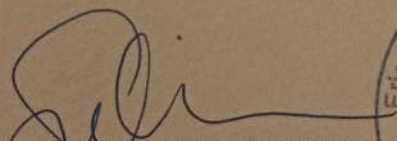
Herrn Juan David Munoz Bolanos
geb. 04.06.1994 in La Cruz (Kolumbien)
den Hochschulgrad

MASTER OF SCIENCE (M.Sc.)

Er hat die Masterprüfung (120 ECTS)
im Studiengang »Photonics«

bestanden.

Jena, 20.12.2021


Prof. Dr. Christian Spielmann
Dekan




Prof. Dr. Silvana Botti
Vorsitzende des
Prüfungsausschusses

OFFICIAL TRANSLATION OF DOCUMENTS WRITTEN IN SPANISH, MADE 20
DECEMBER 2018 BY JUAN CARLOS OJEDA-GARCES, OFFICIAL TRANSLATOR
BY RESOLUTION No. 1171 ISSUED BY THE MINISTRY OF JUSTICE, REPUBLIC
OF COLOMBIA.



UNIVERSIDAD DEL CAUCA
ON BEHALF OF THE
REPUBLIC OF COLOMBIA
AN BY AUTHORIZATION OF THE MINISTRY OF NATIONAL EDUCATION
CONSIDERING THAT

JUAN DAVID MUÑOZ BOLAÑOS
CC 1.061.772.278 OF POPAYAN

HAS COMPLIED WITH ALL THE STATUTORY AND LEGAL REQUIREMENTS,
GRANTS THE TITLE OF

PHYSICAL ENGINEER

WITH ALL THE RIGHTS, PRIVILEGES AND DIGNITIES THAT AUTHORIZE HIM
FOR THE PROFESSIONAL EXERCISE

POPAYAN, 16 MARCH 2018

REGISTERED IN THE DIPLOMA BOOK No. 082 FOLIO 437 DIPLOMA No. 396
RESOLUTION No. R-266-07 MINUTE No. 14-07

(SIGNED)

THE PRINCIPAL OF THE UNIVERSITY
THE DEAN OF THE FACULTY
THE SECRETARY GENERAL OF THE UNIVERSITY

Juan Carlos Ojeda - Garcés
TRADUCTOR INTERPRETE OFICIAL
RESOLUCIÓN No 1171
MINISTERIO DE JUSTICIA 1997

Juan Munoz Bolanos

has participated in the

**ZEISS European
Summer School**

Lithography Optics &
Semiconductor Technologies

26 - 27 June 2025

ZEISS Semiconductor Manufacturing Technology, Oberkochen, Germany

Thomas Marschner

Dr. Thomas Marschner
Program Manager Funding Projects

Thomas Stammer

Dr. Thomas Stammer
CTO & Head of Product Strategy



Zertifikat

Muñoz-Bolaños JUAN DAVID

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Schriftliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden



Datum: 16.09.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



ÖSD ist Mitglied von



ösd

österreich schweiz deutschland

ID-Nummer: ZB2Ö25s34291

Zertifikat

Juan David MUÑOZ BALAÑOS

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Mündliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden



Datum: 11.08.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



ÖSD ist Mitglied von



ösd

österreich schweiz deutschland



ID-Nummer: ZB2Ö25m40434

Passaporto № / Passport No.

MUNOZ BOLANOS

COLOMBIANA

Sexo / Sex
Lugar de nacimiento / Place of birth

Fecha de expedición / Date of issue

Fecha de vencimiento / Date of expiry

G. ANTIOQUIA

Forma dei tubuli / *Peperideris cognominata*

Dear Family

[illegible]

